costing_GATEKEEPER_AUDIT

Gatekeeper audit: final prototype sign-off (2026-11-02)

Scope: all 20 released boards, the generated enclosures (`utility10/pcb/enclosure.py`), the ASTRADAND hilt CAD, the automata gear/cam CAD, the toy manuals, and the live LCSC BOM data. Evidence: tool output and the files themselves; nothing has been built.

Status: HOLD FOR REVISION (portfolio). Bare-PCB bench prototypes are approved per the table below; enclosures and anything for children are not.

Blocking issues (ranked)

1. **MITTI-NETRA: TP4056 exposed pad (U4 pad 9) has no net.** Measured in the released PCB. About 0.8 W at the start of a charge (4.65 V in, 3.2 V cell, 0.55 A) with no thermal path: the IC will thermally throttle and run hot. Fix: tie pad 9 to GND (pour), re-run DRC.
2. **MITTI-NETRA: Li-ion charging outdoors with no temperature cut-off.** TEMP pin is tied to GND (disables the TP4056 NTC function); a sealed box in Indian sun exceeds 45 C, where Li-ion must not be charged. Fix: 10 k NTC divider on TEMP placed against the cell, or switch to LiFePO4 with a matching charger.
3. **All generated enclosures: the lid is held only by a friction lip** (no screws, no snap). It falls off on a drop, and any battery inside is reachable without a tool. Fix: add 2 to 4 lid screw bosses (M2.5 into 2.1 mm pilots) or snap hooks with a defined interference.
4. **All generated enclosures: tact switches are unreachable.** The B3U switch top sits 10.4 to 11.4 mm below the lid surface under a 5 mm hole (12 boards). Fix: printed plunger caps with a retaining flange, or a taller switch (12 x 12 mm tactile with cap).
5. **Toy products (Desi Arcade, Beat Runner): the manuals promise a screw-closed battery door; the CAD has no battery bay at all** (inner height 11 to 12 mm; a 3 x AAA holder is about 14 mm tall). Fix: design a base with a screw-retained battery compartment before any child testing.
6. **KINETIC and MITTI-NETRA enclosures ignore plug-in modules.** StepStick drivers with heatsinks (about 20 mm above the socket) and the SIM7080G breakout plus the 18650 cell are not in the height model; the lid collides, and the driver box has no ventilation.
7. **BioTray / AQUAPONICS: enclosure is not water-resistant** (friction lid, open side cut-outs) for a wet environment with 12 V pumps.
8. **Gas-Suraksha: board not finished and the product is not certified**; Jal-Rakshak and the AP1 audio platform have no board.

Fixed during this review (already in the released files)

- Reversed power diodes: MITTI-NETRA D1/D2, KINETIC D5, BioTray D5 (would have blocked charging or back-fed USB).
- Out-of-stock and low-stock parts replaced with in-stock, same-footprint parts (see `costing/DESIGN_REVIEW_LOG.md`).

Non-blocking findings

- Laser-cut automata gears and cams: kerf is applied in the shrinking direction twice (design reduction plus no outward offset of the cutting path): effective gear backlash about 0.55 mm at module 2 (works, a little loose); cams about 0.2 mm small. Correct the offset sign before the production cut file.
- Passive magnetic buzzers (16 ohm) at 4.2 to 4.5 V exceed the 30 mA rating in continuous tones; add 22 ohm series for loud continuous alarms.
- SK6812MINI-B substitute on ASTRADAND: footprint fit unproven; order 2 boards first.
- Lid indicator holes over 0603 LEDs 10 mm below the lid: add light pipes or the LEDs will be hard to see.
- Generated boxes have sharp 90 degree external edges: add a 1 to 2 mm fillet for hand-held products and toys.
- Single-source and thin stock: BQ25570 (391), ESP32-S3-WROOM-1-N8 at LCSC (774, widely available elsewhere), 12 mm buzzer (140), PESD5V0S1BA (175); MPU-6050 is a mature part from one maker.

Checked and found correct

Diode, TVS, Zener, LED, polarised-capacitor and transistor orientation on all boards; supply and ground pin names of every IC (0 errors, `pincheck.py`); ESP32 pin maps against the firmware headers; crystal-less ATmega boards have matching 8 MHz firmware builds; reverse-polarity PMOS on battery inputs; TVS on external inputs; ESP32 EN RC; I2C pull-ups; decoupling present within 16 mm of every IC (the ESP32 module within 25 mm); DRC 0, ERC 0, netlist match and the Gerber check on all 20 boards; ASTRADAND hilt: 12 screws, battery behind the screwed shell, fit-check PASS on the modelled volumes.

Revision 2 (after the fix round): blocking issues closed

#	Blocking issue (revision 1)	Status
1	MITTI-NETRA TP4056 exposed pad unconnected	**Closed**: pad 9 on GND with three thermal vias; board re-routed, DRC 0, netlist match
2	MITTI-NETRA charging outdoors without a temperature cut-off	**Closed**: R16 6.2 k + 10 k B3950 NTC on TEMP: charging only between about 6 and 44 C
3	Friction-only lids	**Closed**: enclosure generator v2: 4 x M2.5 lid screws into corner columns on every box
4	Unreachable buttons	**Closed**: printed captive plungers for every lid button, light pipes for every lid LED, wall sliders or a top stalk for every slide switch
5	Toys without a battery bay	**Closed**: floor battery bay sized for Keystone 2480 / 2465 / 2469 holders with a 2-screw door (tool needed); arcade consoles with four 30 mm button holes
6	KINETIC / MITTI enclosures ignoring modules and cell	**Closed**: KINETIC box sized for 26 mm driver modules with vents; MITTI-NETRA and BioTray get a printed plate for a bought IP65 box (cell, modem, glands inside)
7	BioTray / Aquaponics not water-resistant	**Closed**: IP65 box with mounting plate
8	Gas-Suraksha, Jal-Rakshak, AP1 without boards	**Gas-Suraksha and Jal-Rakshak closed** (order-ready, DRC 0, ERC 0, all parts LCSC); **AP1**: see SIGNOFF.md

Further fixes in this round: automata kerf compensation corrected; tilt-maze gimbal drawn; racer and cricket cut files kerf-compensated; Drop Quest v1 rules with 60 solver-verified levels and parts; gravity-toy drawings completed; paper rocket changed to a stomp launcher (no stored pressure); Katha Cubes word fixes; AP1 ball-lane hardware added and all three AP1 apps built for the board; active-buzzer flyback placed across the buzzer on Jal-Rakshak; +4V5 power symbol separated from +5V in schematics.

Remaining conditions are production gates (native language review, lab tests, real calibration, ESP32 compile on target), listed per project in `costing/SIGNOFF.md`.